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FILM FORMING DEVICE AND METHOD FOR FORMING CARBON FILM

Abstract

A film forming device includes: a film forming chamber; a holder configured to hold a substrate in the film forming chamber; an inlet pipe configured to introduce a gas of a raw material containing carbon into the film forming chamber; a cathode electrode that is filamentous; a first power source configured to heat the cathode electrode by energization; an anode electrode provided around the cathode electrode; a second power source configured to generate a discharge between the cathode electrode and the anode electrode; a third power source configured to generate a potential difference between the cathode electrode or the anode electrode and the substrate; an ionization area configured to ionize the gas to generate an ionized gas by the discharge; and an acceleration area in which the ionized gas is accelerated by the potential difference. A soft-magnetic cylinder is provided around the acceleration area.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on and claims priority to Japanese Patent Application No. 2024-033062 filed on Mar. 5, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Field of the Invention

[0002] The present disclosure relates to a film forming device and a method for forming a carbon film.

2. Description of the Related Art

[0003] In recent years, in the field of magnetic recording media used for hard disk drives (HDDs) and the like, enhancement of recording density has been demanded. There are various technologies to support such enhancement of recording density. One of them is to shorten a distance between a magnetic head and a magnetic layer of a magnetic recording medium.

[0004] Here, a protective film is interposed between the magnetic layer of the magnetic recording medium and the magnetic head, which is an issue in reducing the distance between the magnetic head and the magnetic layer. In order to increase the recording density of the magnetic recording medium by shortening the distance between the magnetic head and the magnetic layer, it is necessary to enhance smoothness of the protective film and reduce thickness of the protective film.

[0005] For example, Japanese Laid-Open Patent Application No. 2011-65714 discloses a carbon-film forming device in which the distance between the magnetic recording medium and the magnetic head can be shortened by stabilizing the thickness distribution of the carbon film used as the protective film of the magnetic recording medium.

SUMMARY

[0006] The present disclosure includes the following:

[0007] [1] A film forming device, comprising: [0008] a film forming chamber; [0009] a holder configured to hold a substrate in the film forming chamber; [0010] an inlet pipe configured to introduce a gas of a raw material containing carbon into the film forming chamber; [0011] a cathode electrode that is filamentous; [0012] a first power source configured to heat the cathode electrode by energization; [0013] an anode electrode provided around the cathode electrode; [0014] a second power source configured to generate a discharge between the cathode electrode and the anode electrode; [0015] a third power source configured to generate a potential difference between the cathode electrode or the anode electrode and the substrate; [0016] an ionization area configured to ionize the gas to generate an ionized gas by the discharge; and [0017] an acceleration area in which the ionized gas is accelerated by the potential difference, wherein [0018] a soft-magnetic cylinder is provided around the acceleration area.

[0019] [2] The film forming device according to [1], wherein [0020] the soft-magnetic cylinder has a saturation magnetic-flux density of 0.5 T or more.

[0021] [3] The film forming device according to [1] or [2], wherein [0022] the soft-magnetic cylinder has a coercivity of 0.5 A/m or less.

[0023] [4] The film forming device according to any one of [1] to [3], wherein [0024] the soft-magnetic cylinder is made of an amorphous metal containing one or more elements selected from a group consisting of Co, Fe, and Ni.

[0025] [5] A method for forming a carbon film on a surface of a substrate, the method using a film forming device including: [0026] a film forming chamber; [0027] a holder configured to hold a

substrate in the film forming chamber; [0028] an inlet pipe configured to introduce a gas of a raw material containing carbon into the film forming chamber; [0029] a cathode electrode that is filamentous; [0030] a first power source configured to heat the cathode electrode by energization; [0031] an anode electrode provided around the cathode electrode; [0032] a second power source configured to generate a discharge between the cathode electrode and the anode electrode; [0033] a third power source configured to generate a potential difference between the cathode electrode or the anode electrode and the substrate; [0034] an ionization area configured to ionize the gas to generate an ionized gas by the discharge; and [0035] an acceleration area in which the ionized gas is accelerated by the potential difference, wherein [0036] the film forming device is provided with a soft-magnetic cylinder provided around the acceleration area, the method comprising: [0037] introducing the gas into the film forming chamber from the inlet pipe; [0038] ionizing the gas by the discharge between the cathode electrode that is filamentous and heated by energization and the anode electrode provided around the cathode electrode; and [0039] emitting the ionized gas toward the surface of the substrate by accelerating the ionized gas inside the acceleration area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] FIG. 1 is a schematic structural diagram schematically illustrating a film forming device according to an embodiment of the present disclosure;

[0041] FIG. 2 is a cross-sectional view illustrating an example of a magnetic recording medium manufactured by applying the film forming device according to the embodiment of the present disclosure;

[0042] FIG. 3 is a perspective view illustrating an example of a magnetic recording and reproducing device;

[0043] FIG. 4 is a plan view illustrating a configuration of an in-line film forming device to which the film forming device according to the embodiment of the present disclosure is applied;

[0044] FIG. 5 is a side view illustrating carriers of an in-line film forming device to which the film forming device according to the embodiment of the present disclosure is applied; and

[0045] FIG. 6 is an enlarged side view of the carrier as illustrated in FIG. 5.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0046] Demand for enhancement of recording density of magnetic recording media continues, and a film forming device and a method for forming a carbon film which can further smooth a protective film are required.

[0047] The present disclosure has been made in view of the above circumstances, and an object of the present disclosure is to provide a film forming device and a method for forming a carbon film which can stabilize the thickness distribution of the carbon film.

[0048] An embodiment of the present disclosure will now be described in detail with reference to the drawings. In the drawings used in the following description, the characteristic parts are enlarged for convenience, and dimensional proportions of components are not necessarily the same as in reality. In this specification, “to” indicating a numerical range means that the numerical values described before and after it are included as the lower limit and upper limit, unless otherwise specified. In the numerical range represented by “to”, when only the upper limit is indicated in units, the lower limit is indicated in the same units.

(Film Forming Device and Method for Forming Carbon Film)

[0049] The film forming device and the method for forming a carbon film according to the embodiment of the present disclosure will be described in the following. In the present embodiment, the film forming device according to the present embodiment is a carbon-film forming device for forming a carbon film as the protective film.

[0050] FIG. 1 is a schematic structural diagram schematically illustrating the film forming device according to the present embodiment. As illustrated in FIG. 1, a film forming device **100** is a film forming device that uses an ion-beam deposition method. The film forming device **100** includes a film forming chamber **101**, a holder **102** for holding a substrate D in the film forming chamber **101**, an inlet pipe **103** for introducing a gas G of a carbon-containing raw material into the film forming chamber **101**, a filamentous cathode electrode **104** arranged in the film forming chamber **101**, and an anode electrode **105** arranged around the cathode electrode **104** in the film forming chamber **101**. The film forming device **100** includes a first power source **106** for heating the cathode electrode **104** by energization, a second power source **107** for generating a discharge between the cathode electrode **104** and the anode electrode **105**, and a third power source **108** for providing a potential difference between the cathode electrode **104** or the anode electrode **105** and the substrate D. Furthermore, the film forming device **100** includes an ionization area **111** for ionizing a gas by the discharge, an acceleration area **112** for accelerating the ionized gas by the potential difference, and a soft-magnetic cylinder **109** provided around the acceleration area **112**.

[0051] The term “gas G of the carbon-containing raw material” means that the carbon-containing raw material is a gas, and is a raw-material gas obtained by gasifying the carbon-containing raw material.

[0052] The film forming chamber **101** is air-tight with a chamber wall **101a**, and is configured such that the inside can be evacuated under reduced pressure through an exhaust pipe **110** connected to a vacuum pump (not illustrated).

[0053] The first power source **106** is an AC power source connected to the cathode electrode **104**, and supplies power to the cathode electrode **104** when forming a carbon film. The first power source **106** is not limited to the AC power source, and may be a DC power source.

[0054] The second power source **107** is a DC power source connected to the cathode electrode **104** on a negative electrode side and to the anode electrode **105** on a positive electrode side, and causes a discharge between the cathode electrode **104** and the anode electrode **105** when forming a carbon film.

[0055] The third power source **108** is a DC power source in which the positive electrode side is connected to the anode electrode **105** and the negative electrode side is connected to the holder **102**, and imparts a potential difference between the anode electrode **105** and the substrate D held by the holder **102** when forming a carbon film. The third power source **108** may have a configuration in which the positive electrode side is connected to the cathode electrode **104**.

[0056] In the film forming chamber **101**, the ionization area **111** in which gas is ionized by a discharge and the acceleration area **112** in which the ionized gas is accelerated by a potential difference are formed. The soft-magnetic cylinder **109** is provided around the acceleration area **112**.

[0057] According to an examination by the inventors of the present disclosure, it was noticed that in an existing film forming device, no drastic measures were taken against magnetic influences such as an external leakage magnetic field, and that this was a factor that made the film thickness distribution of the carbon film unstable. That is, there are often magnets for a magnetron of a sputtering device or magnets used for a transfer unit of a processing substrate around the film forming device, and the leakage magnetic field from these magnets adversely affects the film forming device. The inventors of the present disclosure found that when the soft-magnetic cylinder **109** is provided around the acceleration area **112**, the film thickness distribution of the carbon film to be formed can be stabilized.

[0058] In the present embodiment, by providing the soft-magnetic cylinder **109** around the acceleration area **112**, paths of ions accelerated in a direction of the substrate D (right direction in FIG. 1) are prevented from being disturbed by the leakage magnetic field, and a carbon film having a substantially uniform film thickness can be formed on one surface of the substrate D (film formation surface).

[0059] The soft-magnetic cylinder **109** is preferably rotationally symmetric about a central axis

perpendicular to the film formation surface of the substrate D. In this manner, the acceleration area **112** can be rotationally symmetric about the film formation surface of the substrate D, such that the film thickness distribution of the carbon film on the film formation surface can be made more uniform.

[0060] The soft-magnetic cylinder **109** is preferably in an electrically floating state (floating potential). In this way, the potential distribution in the soft-magnetic cylinder **109** becomes more uniform, and the paths of the ions accelerated in the soft-magnetic cylinder **109** become more stable and perpendicular to the film formation surface of the substrate D.

[0061] A saturation magnetic-flux density B_s of the soft-magnetic cylinder **109** is 0.5 Tesla (T) or more and preferably as high as possible. Accordingly, the influence of the leakage magnetic field in the soft-magnetic cylinder **109** can be further reduced.

[0062] A coercivity H_c of the soft-magnetic cylinder **109** is preferably 0.5 Ampere per meter (A/m) or less and as low as possible. In this manner, it is possible to suppress magnetization of the soft-magnetic cylinder **109** by the leakage magnetic field, and thus, influence of the magnetic field in the soft-magnetic cylinder **109** can be further reduced.

[0063] The soft-magnetic cylinder **109** is preferably an amorphous metal containing one or more elements selected from a group consisting of Co, Fe, and Ni. In this manner, the influence of the leakage magnetic field in the soft-magnetic cylinder **109** can be further reduced, and magnetization of the soft-magnetic cylinder **109** by the leakage magnetic field can be further suppressed.

[0064] In the film forming device **100**, when a carbon film is formed on the substrate D, the voltage and current of the first power source **106**, the second power source **107**, and the third power source **108** may be appropriately set according to the size of the substrate D. When the substrate D is, for example, a disk-shaped substrate having an outer diameter of 3.5 inches, the voltage of the first power source **106** is preferably set within a range from 10 to 100 V and the current is preferably set within a range from 5 to 50 A in direct current or alternating current. The voltage of the second power source **107** is preferably set within a range from 50 to 300 V and the current is preferably set within a range from 10 to 5000 mA. The voltage of the third power source **108** is preferably set within a range from 30 to 500 V and the current is preferably set within a range from 10 to 200 mA.

[0065] When a carbon film is formed on the surface of the substrate D by using the film forming device **100** having the above-described configuration, the gas G of the carbon-containing raw material is introduced into the film forming chamber **101** which has been reduced in pressure through the exhaust pipe **110** through the inlet pipe **103**. The raw-material gas G is excited and decomposed into ionized gas (carbon ions) by thermal plasma of the cathode electrode **104** heated by the supply of electric power from the first power source **106** and plasma in the ionization area **111** generated by the discharge between the cathode electrode **104** and the anode electrode **105** connected to the second power source **107**. The carbon ions excited by the plasma in the ionization area **111** collide with the surface of the substrate D while accelerating in the acceleration area **112** toward the substrate D which is set to a negative potential by the third power source **108**.

[0066] In the method for forming a carbon film by using the film forming device **100**, for example, a hydrocarbon gas can be used as the gas G of the carbon-containing raw material. As the hydrocarbon, it is preferable to use one or more low-carbon hydrocarbons selected from a group consisting of lower saturated hydrocarbons, lower unsaturated hydrocarbons, and lower cyclic hydrocarbons. In this case, "low carbon" means that the number of contained carbons is within a range from 1 to 10.

[0067] As the lower saturated hydrocarbon, methane, ethane, propane, butane, octane, and the like can be used.

[0068] As the lower unsaturated hydrocarbon, isoprene, ethylene, propylene, butylene, butadiene, and the like can be used.

[0069] As the lower cyclic hydrocarbon, benzene, toluene, xylene, styrene, naphthalene,

cyclohexane, cyclohexadiene, and the like can be used.

[0070] The reason why it is preferable to use the lower hydrocarbons is that, when the number of carbons in the hydrocarbon exceeds the range specified above, it becomes difficult to supply the hydrocarbon as the gas from the inlet pipe **103**, decomposition of the hydrocarbon during discharge becomes difficult, and a generated carbon film thus contains a large amount of polymer components which are inferior in strength.

[0071] In order to induce generation of the plasma in the film forming chamber **101**, it is preferable to use a mixture gas, which is the gas obtained by mixing inert gas, hydrogen gas, and the like with the gas G of the carbon-containing raw material. A mixing ratio of the hydrocarbon and the inert gas in the mixture gas is, in a ratio of the hydrocarbon to the inert gas, preferably set within a range from 2:1 to 1:100 (volume ratio). In this way, a carbon film with high hardness and high durability can be formed.

[0072] In the film forming device **100**, a carbon film is formed only on one surface (one side) of the substrate D, but it may be formed on both surfaces (both sides) of the substrate D. In this case, the same device configuration as in the case where a carbon film is formed only on one surface of the substrate D may be arranged on both sides of the substrate D in the film forming chamber **101**. (Method of Manufacturing Magnetic Recording Medium)

[0073] Next, a method of manufacturing a magnetic recording medium by applying the film forming device **100** will be described. In the present embodiment, a case of manufacturing a magnetic recording medium to be mounted on a hard disk device will be described by using an in-line film forming device in which film-forming processes are performed while substrates on which a carbon film is to be formed are sequentially conveyed between a plurality of film forming chambers.

(Magnetic Recording Medium)

[0074] As illustrated in FIG. 2, for example, a magnetic recording medium **200** manufactured by using the film forming device **100** has a structure in which a magnetic layer **220**, a protective layer **230**, and a lubricant film **240** are stacked in order on both surfaces of a non-magnetic substrate **210**. The magnetic layer **220** includes a soft-magnetic layer **221**, an intermediate layer **222**, and a recording magnetic layer **223**, and has a stacking structure where the layers are stacked in this order from a side of the non-magnetic substrate **210**.

[0075] In the magnetic recording medium **200**, a carbon film with high hardness and high density is formed with a uniform thickness as the protective layer **230** by using the film forming device **100**. In this case, the thickness of the carbon film in the magnetic recording medium **200** can be made thin, for example, the thickness of the carbon film can be made about 2 nm or less.

[0076] Therefore, since a distance between the magnetic recording medium **200** and the magnetic head can be shortened through use of the film forming device **100**, a recording density of the magnetic recording medium **200** can be enhanced and the corrosion resistance of the magnetic recording medium **200** can be enhanced.

[0077] Each layer of the magnetic recording medium **200** other than the protective layer **230** will be described in the following.

[0078] As the non-magnetic substrate **210**, any non-magnetic substrate, for example, a substrate including an Al alloy formed mainly of Al such as an Al—Mg alloy and the like, an ordinary soda glass, an aluminosilicate glass, a crystallized glass, silicon, titanium, ceramics, and various resins can be used. Among these, it is preferable to use an Al alloy substrate made of an Al alloy, a glass substrate made of crystallized glass, or a silicon substrate made of silicon. An average surface roughness (Ra) in each of these substrates is preferably 1 nm or less, more preferably 0.5 nm or less, and more preferably 0.1 nm or less.

[0079] The magnetic layer **220** may be an in-plane magnetic layer for an in-plane magnetic recording medium or a perpendicular magnetic layer for a perpendicular magnetic recording medium, but it is preferable to use the perpendicular magnetic layer in order to realize a higher

recording density. The magnetic layer **220** is preferably formed of an alloy mainly formed of Co. For example, as the magnetic layer **220** for the perpendicular magnetic recording medium, for example, a laminate of the soft-magnetic layer **221** made of a soft-magnetic FeCo alloy (FeCoB, FeCoSiB, FeCoZr, FeCoZrB, FeCoZrBCu, etc.), a FeTa alloy (FeTaN, FeTaC, etc.), or a Co alloy (CoTaZr, CoZrNB, CoB, etc.), the intermediate layer **222** made of Ru or the like, and the recording magnetic layer **223** made of 60Co-15Cr-15 Pt alloy or 70Co-5Cr-15 Pt-10SiO.sub.2 alloy can be used.

[0080] The overall thickness of the magnetic layer **220** is within a range from 3 to 20 nm, preferably within a range from 5 to 15 nm, and the magnetic layer **220** may be formed to obtain sufficient head input and output according to the type of magnetic alloy to be used and the stacking structure. For the thickness of the magnetic layer **220**, although it is necessary to have a certain thickness of the magnetic layer in order to obtain a certain level of output during reproduction, because parameters representing recording and reproduction characteristics usually deteriorate as the output increases, it is necessary to set an optimum film thickness.

[0081] As a lubricant used for the lubricant film **240**, a fluorinated liquid lubricant such as perfluoroether (PFPE), or a solid lubricant such as a fatty acid can be used. The lubricant film **240** is usually formed with a thickness within a range from 1 to 4 nm. As a lubricant coating method, a conventionally known general coating method, such as a dipping method or a spin coating method, can be used.

(Magnetic Recording and Reproducing Device)

[0082] As a magnetic recording and reproducing device using the magnetic recording medium **200**, a hard disk device can be cited, for example. As illustrated in FIG. 3, a hard disk device **300** includes a magnetic disk **310** formed of the magnetic recording medium **200** (see FIG. 2), a medium driver **320** for rotating and driving the magnetic disk **310**, a magnetic head **330** for recording and reproducing information on the magnetic disk **310**, a head driver **340**, and a recording-and-reproducing signal processing system **350**. The recording-and-reproducing signal processing system **350** processes input data, sends a recording signal to the magnetic head **330**, and processes a reproduction signal from the magnetic head **330** to output data.

(In-Line Film Forming Device) In manufacturing the magnetic recording medium **200**, an in-line film forming device (manufacturing device for magnetic recording medium) as illustrated in FIG. 4, for example, to which the film forming device **100** is applied is used. By using the in-line film forming device as illustrated in FIG. 4, the magnetic layer **220** including at least the soft-magnetic layer **221**, the intermediate layer **222**, and the recording magnetic layer **223**, the protective layer **230**, and the lubricant film **240** are successively laminated on both surfaces of the non-magnetic substrate **210**, which is the target of the film formation, and thus, the above-mentioned magnetic recording medium **200** having a high hardness and dense carbon film as the protective layer **230** can be stably manufactured.

[0083] Specifically, the above-mentioned in-line film forming device includes a stand for a cassette-substrate transfer robot (robot stand) **1**, a substrate-supply robot chamber **2** adjacent to the robot stand **1**, a cassette-substrate transfer robot **3** arranged on the robot stand **1**, a substrate-supply robot **34** arranged in the substrate-supply robot chamber **2**, and a substrate mounting chamber **52** adjacent to the substrate-supply robot chamber **2**. Furthermore, the above-mentioned in-line film forming device includes corner chambers **4**, **7**, **14**, and **17** for rotating a carrier **25**, processing chambers **5**, **6**, **8** to **13**, **15**, **16**, and **18** to **20** arranged between the corresponding corner chambers **4**, **7**, **14**, and **17**, a substrate removal chamber **54** arranged adjacent to the processing chamber **20**, and an ashing chamber **3A** arranged between the substrate mounting chamber **52** and the substrate removal chamber **54**. The above-mentioned in-line film forming device includes a substrate-removal robot chamber **22** arranged adjacent to the ashing chamber **3A**, a substrate-removal robot **49** arranged in the substrate-removal robot chamber **22**, and a plurality of carriers **25** conveyed through each of these chambers **2**, **3A**, **4** to **20**, **52**, and **54**.

[0084] The term “chamber” means any one of the substrate-supply robot chamber **2**, the ashing chamber **3A**, the corner chambers **4**, **7**, **14**, and **17**, the processing chambers **5**, **6**, **8** to **13**, **15**, **16**, and **18** to **20**, the substrate mounting chamber **52**, and the substrate removal chamber **54**.

[0085] Each of the chambers **2**, **3A**, **4** to **20**, **52**, and **54** is connected to two adjacent walls, and gate valves **55** to **72** are provided at the corresponding connection portion of the chambers **2**, **3A**, **4** to **20**, **52**, and **54**. When the gate valves **55** to **72** are closed, each of the chambers **2**, **3A**, **4** to **20**, **52**, and **54** becomes an independent enclosed space.

[0086] A vacuum pump (not illustrated) is connected to each of the chambers **2**, **3A**, **4** to **20**, **52**, and **54**. The magnetic recording medium **200** as illustrated in FIG. **2** is finally obtained by successively forming the magnetic layer **220** (the soft-magnetic layer **221**, the intermediate layer **222**, and the recording magnetic layer **223**) and the protective layer **230** on both surfaces of the non-magnetic substrate **210** (see FIG. **2**) mounted on the carrier **25** in the chambers **2**, **3A**, **4** to **20**, **52**, and **54**, while the carrier **25** is successively conveyed by a conveying mechanism (not illustrated) into each of the chambers **2**, **3A**, **4** to **20**, **52**, and **54**, which are reduced in pressure by the operation of the vacuum pumps.

[0087] Among the chambers **2**, **3A**, **4** to **20**, **52**, and **54**, the corner chambers **4**, **7**, **14**, and **17** are chambers for changing a moving direction of the carrier **25**, and a mechanism for rotating the carrier **25** to move it to the next film forming chamber is provided in the corner chambers **4**, **7**, **14**, and **17**.

[0088] The cassette-substrate transfer robot **3** supplies the non-magnetic substrate **210** to the substrate-supply robot chamber **2** from the cassette containing the non-magnetic substrate **210** before film formation. Additionally, the cassette-substrate transfer robot **3** retrieves the non-magnetic substrate **210** (magnetic recording medium **200**) after film formation removed by the substrate-removal robot **49**, from the substrate-removal robot chamber **22**. The substrate-supply robot chamber **2** is provided, at one side wall of the chamber, with an opening **51A** open to the outside and a gate valve **31A** for opening and closing the opening **51A**. The substrate-removal robot chamber **22** is provided, at one side wall of the chamber, with an opening **51B** open to the outside and a gate valve **31B** for opening and closing the opening **51B**.

[0089] Inside the substrate mounting chamber **52**, the non-magnetic substrate **210** before film formation is mounted on the carrier **25** by using the substrate-supply robot **34**. Additionally, inside the substrate removal chamber **54**, the non-magnetic substrate **210** (magnetic recording medium **200**) after film formation mounted on the carrier **25** is removed by using the substrate-removal robot **49**.

[0090] The ashing chamber **3A** performs ashing of the carrier **25** conveyed from the substrate removal chamber **54**, and then conveys the carrier **25** to the substrate mounting chamber **52**.

[0091] Among the processing chambers **5**, **6**, **8** to **13**, **15**, **16**, and **18** to **20**, the processing chambers **5**, **6**, **8** to **13**, **15**, and **16** constitute a plurality of film forming chambers for forming the magnetic layer **220**. These film forming chambers include mechanisms for forming the soft-magnetic layer **221**, the intermediate layer **222**, and the recording magnetic layer **223** on both sides of the non-magnetic substrate **210**.

[0092] Furthermore, the processing chambers **18** to **20** constitute a plurality of film forming chambers for forming the protective layer **230**. These film forming chambers have the same device configuration as the film forming device using the ion-beam deposition method as illustrated in FIG. **1**, and form the dense carbon film with high hardness described above as the protective layer **230**, on top of the magnetic layer **220** formed on both surfaces of the non-magnetic substrate **210**.

[0093] Each of the processing chambers **5**, **6**, **8** to **13**, **15**, **16**, and **18** to **20** includes two processors (a pair of processors **41a** and **41b**, a pair of processors **45a** and **45b**, or a pair of processors **48a** and **48b**), one on each of the two chamber sides facing the carrier **25**. One of the pair of processors, that is, the processor **41a**, **45a**, or **48a** performs film formation on the left side of the non-magnetic substrate **210** with respect to a traveling direction of the carrier **25**, and the other of the pair of

processors, that is, the processor **41b**, **45b**, or **48b** performs film formation on the right side of the non-magnetic substrate **210** with respect to the traveling direction of the carrier **25**.

[0094] Each of the processing chambers **5**, **6**, **8** to **13**, **15**, **16**, and **18** to **20** is provided with a processing-gas supply pipe. The processing-gas supply pipe is provided with a valve whose opening and closing is controlled by a control mechanism (not illustrated). By opening and closing the valve of each processing-gas supply pipe and a pump gate valve, the supply of gas from the processing-gas supply pipe and the pressure and discharge of the gas in the chamber are controlled.

[0095] As illustrated in FIGS. **5** and **6**, the carrier **25** includes a support base **26** and a plurality of substrate mounting portions **27** provided on an upper surface of the support base **26**. In the present embodiment, since two substrate mounting portions **27** are mounted, two non-magnetic substrates **210** mounted on the substrate mounting portions **27** are hereinafter treated as a first film-formation substrate **23** and a second film-formation substrate **24**, respectively.

[0096] The substrate mounting portion **27** includes a plate **28** having a thickness of about 1 to several times the thickness of the first film-formation substrate **23** and the second film-formation substrate **24**, a circular through-hole **29** formed slightly larger in diameter than an outer periphery of the first film-formation substrate **23** and the second film-formation substrate **24**, and a plurality of supports **30** projecting toward the inside of the through-hole **29** around the through-hole **29**. In the substrate mounting portion **27**, the through-hole **29** is formed in the plate **28**, and a plurality of supports **30** projecting toward the inside of the through-hole **29** are provided on the periphery of the through-hole **29**. In the substrate mounting portion **27**, the first film-formation substrate **23** and the second film-formation substrate **24** are respectively fitted into the corresponding through-hole **29**, and the supports **30** are engaged with an outer edge of the fitted substrate, such that the first film-formation substrate **23** and the second film-formation substrate **24** are held vertically (in a state where the main surfaces of each of the first film-formation substrate **23** and the second film-formation substrate **24** are parallel to a gravity direction). That is, the substrate mounting portion **27** is arranged on the upper surface of the support base **26** while being aligned with the other substrate mounting portion **27**, such that the main surfaces of the first film-formation substrate **23** and the second film-formation substrate **24** mounted on the carrier **25** are substantially orthogonal to the upper surface of the support base **26**, and the main surfaces of the first film-formation substrate **23** and the second film-formation substrate **24** are on the same plane.

[0097] Moreover, each of the above-described processing chambers **5**, **6**, **8** to **13**, **15**, **16**, and **18** to **20** is provided with the two processors (the pair of processors **41a** and **41b**, the pair of processors **45a** and **45b**, or the pair of processors **48a** and **48b** in FIG. **4**) one on each of the two chamber sides facing the carrier **25**. In this case, for example, in a state where the carrier **25** is stopped at a first processing position indicated by a solid line in FIG. **5**, film-forming processing is performed on the first film-formation substrate **23** on the left side of the carrier **25**. Thereafter, the carrier **25** moves to a second processing position indicated by a dashed line in FIG. **5**, and in a state where the carrier **25** is stopped at the second processing position, the film-forming processing is performed on the second film-formation substrate **24** on the right side of the carrier **25**.

[0098] In a case where there are a total of four processors on two chamber sides facing the carrier **25**, that is the pair of facing processors being respectively provided to face the main surfaces of each of the first film-formation substrate **23** and the second film-formation substrate **24**, it is not necessary to move the carrier **25**, and film-forming processing can be simultaneously performed on both the first film-formation substrate **23** and the second film-formation substrate **24** held by the carrier **25**.

[0099] After the film formation, the first film-formation substrate **23** and the second film-formation substrate **24** are removed from the carrier **25**, and only the carrier **25** on which a carbon film is formed is conveyed into the ashing chamber **3A** as illustrated in FIG. **1**. Then, oxygen gas is introduced from any part of the ashing chamber **3A**, and oxygen plasma is generated in the ashing chamber **3A** by using the oxygen gas. When the oxygen plasma contacts the carbon film formed on

the surface of the carrier 25, the carbon film is decomposed into CO or CO₂ gas and removed.

EXAMPLES

[0100] Hereinafter, the present embodiment will be specifically described based on the Examples, but the present embodiment is not limited to these Examples.

<Manufacturing of Magnetic Recording Medium>

Example 1

[0101] In Example 1, an aluminum substrate on which NiP plating was applied was prepared as a non-magnetic substrate. Next, a magnetic layer was formed by successively laminating a soft-magnetic layer formed of FeCoB with a film thickness of 60 nm, an intermediate layer formed of Ru with a film thickness of 10 nm, and a recording magnetic layer formed of 70Co-5Cr-15Pt-10SiO₂ alloy with a film thickness of 15 nm on both sides of the non-magnetic substrate mounted on an A5052 aluminum alloy carrier by using the in-line film forming device as illustrated in FIG. 4. Next, the non-magnetic substrate mounted on the carrier was transported to a processing chamber having the same device configuration as that of the film forming device as illustrated in FIG. 1, and a protective layer formed of a carbon film was formed on both sides of the non-magnetic substrate on which the magnetic layers were formed.

[0102] Specifically, the processing chamber of the carbon-film forming device had a cylindrical shape with an outer diameter of 180 mm and a length of 250 mm. The chamber wall constituting the processing chamber was made of SUS304. A coiled cathode electrode made of tantalum of about 30 mm in length and a cylindrical anode electrode surrounding the cathode electrode were provided in the processing chamber. The anode electrode was made of SUS304, with an outer diameter of 140 mm and a length of 40 mm. The distance between the cathode electrode and the non-magnetic substrate was 160 mm. A soft-magnetic cylinder was provided between the cathode electrode and the anode electrode and the non-magnetic substrate. The cylinder had a floating potential. The cylinder had an outer diameter of 150 mm and a length of 130 mm. The composition of the cylinder was amorphous 70Ni-10Co-10Fe-5Si-5B (at %). The saturation magnetic-flux density B_s was 0.6 T and the coercivity H_c was 0.2 A/m.

[0103] Gasified toluene was used as the raw-material gas. As for film-forming conditions of the carbon film, a gas flow rate was 2.2 SCCM (1 SCCM of Ar as the carrier gas), the reaction pressure was 0.2 Pa, the cathode voltage was 110 V, the voltage between the cathode electrode and the anode electrode was 75 V, the current was 1300 mA, the ion acceleration voltage was 100 V at 180 mA, the length of film formation time was 2 seconds, and the thickness of the carbon film to be formed was 1.5 nm.

Examples 2 and 3 and Comparative Examples 1 to 3

[0104] A magnetic recording medium was manufactured under the same conditions as in Example 1, but the cylinder in the carbon-film forming device was changed to the contents as illustrated in Table 1. In Comparative Example 3, the cylinder was not provided, and a cylindrical magnet was provided outside the outer wall of the acceleration area of the carbon-film forming device. The magnet had a shape of 185 mm in inner diameter and 40 mm in length, and 20 NdFe-based sintered rod magnets of 10 mm square side lengths and 40 mm in length were arranged in parallel at equal intervals. The sintered rod magnets were arranged such that the S pole was on the substrate side and the N pole was on the cathode side. The total magnetic force of this magnet was 50 G (5 mT).

<Evaluation of Magnetic Recording Media>

[0105] Corrosion tests were performed on the magnetic recording media of each Example and Comparative Example.

(Corrosion Test)

[0106] In the corrosion test, after the magnetic recording media was left at 90° C. and 90% humidity for 96 hours, the amounts of Co elution on the outer-peripheral, center-peripheral, and inner-peripheral surfaces of the magnetic recording media were measured. When the carbon film thickness of the magnetic recording media is uneven, Co contained in the magnetic layer elutes to

the surface due to corrosion. Therefore, the unevenness of the carbon film thickness can be evaluated by measuring the amount of Co elution on the surface of the magnetic recording medium. In the evaluation, the larger the amount of Co elution, the more uneven the carbon film thickness. The measurement results are as illustrated in Table 1.

TABLE-US-00001 TABLE 1 Co ELUTION AMOUNT [$\mu\text{g}/\text{m}^2$] CYLINDER INNER-CENTER- OUTER- Bs Hc PERIPHERAL PERIPHERAL PERIPHERAL MATERIAL [at %]
CHARACTERISTICS [T] [A/m] SURFACE SURFACE SURFACE EXAMPLE 1 70Ni—10Co—10Fe—5Si—5B SOFT-MAGNETIC, 0.6 0.2 0.2 0.5 1.0 AMORPHOUS EXAMPLE 2 78Ni—12Fe SOFT-MAGNETIC, 0.7 1.0 0.25 0.7 1.5 POLYCRYSTALLINE EXAMPLE 3 SUS430 SOFT-MAGNETIC, 1.1 2.4 0.3 0.8 1.8 (83Fe—17Cr) POLYCRYSTALLINE COMPARATIVE ALUMINUM NON-MAGNETIC, 0 0 0.3 1.0 2.0 EXAMPLE 1 POLYCRYSTALLINE COMPARATIVE SUS304 NON-MAGNETIC, 0 0 0.3 1.0 2.0 EXAMPLE 2 (74Fe—18Cr—8Ni) POLYCRYSTALLINE COMPARATIVE NOT PROVIDED, (APPLIED MAGNETIC 0.3 0.9 1.9 EXAMPLE 3 FIELD EXTERNALLY)

[0107] According to Table 1, the outer-peripheral, center-peripheral, and inner-peripheral surfaces of the magnetic recording medium in each example had an overall smaller amount of Co elution than the outer-peripheral, center-peripheral, and inner-peripheral surfaces of the magnetic recording medium of each comparative example. Therefore, it can be said that the carbon films of the magnetic recording media of the examples have a more uniform film thickness than the carbon films of the magnetic recording media of the comparative examples.

[0108] Therefore, it can be said that the film thickness distribution of the carbon film can be stabilized by forming the carbon film as the protective layer on both sides of the non-magnetic substrate on which the magnetic layer is respectively formed, by using the film forming device of the present embodiment. Since the film thickness of the carbon film can be made thin by using the film forming device of the present embodiment, the distance between the magnetic recording medium and the magnetic head can be shortened, and thus, a magnetic recording medium with high recording density can be obtained.

[0109] According to the present disclosure, the thickness distribution of a carbon film can be stabilized. When an obtained carbon film is used as a protective film of a magnetic recording medium or the like, the thickness of the carbon film can be reduced, such that the distance between the magnetic recording medium and the magnetic head can be shortened. As a result, the recording density of the magnetic recording medium can be enhanced.

Claims

1. A film forming device, comprising: a film forming chamber; a holder configured to hold a substrate in the film forming chamber; an inlet pipe configured to introduce a gas of a raw material containing carbon into the film forming chamber; a cathode electrode that is filamentous; a first power source configured to heat the cathode electrode by energization; an anode electrode provided around the cathode electrode; a second power source configured to generate a discharge between the cathode electrode and the anode electrode; a third power source configured to generate a potential difference between the cathode electrode or the anode electrode and the substrate; an ionization area configured to ionize the gas to generate an ionized gas by the discharge; and an acceleration area in which the ionized gas is accelerated by the potential difference, wherein a soft-magnetic cylinder is provided around the acceleration area.
2. The film forming device according to claim 1, wherein the soft-magnetic cylinder has a saturation magnetic-flux density of 0.5 T or more.
3. The film forming device according to claim 1, wherein the soft-magnetic cylinder has a coercivity of 0.5 A/m or less.
4. The film forming device according to claim 1, wherein the soft-magnetic cylinder is made of an

amorphous metal containing one or more elements selected from a group consisting of Co, Fe, and Ni.

5. A method for forming a carbon film on a surface of a substrate, the method using a film forming device including: a film forming chamber; a holder configured to hold a substrate in the film forming chamber; an inlet pipe configured to introduce a gas of a raw material containing carbon into the film forming chamber; a cathode electrode that is filamentous; a first power source configured to heat the cathode electrode by energization; an anode electrode provided around the cathode electrode; a second power source configured to generate a discharge between the cathode electrode and the anode electrode; a third power source configured to generate a potential difference between the cathode electrode or the anode electrode and the substrate; an ionization area configured to ionize the gas to generate an ionized gas by the discharge; and an acceleration area in which the ionized gas is accelerated by the potential difference, wherein the film forming device is provided with a soft-magnetic cylinder provided around the acceleration area, the method comprising: introducing the gas into the film forming chamber from the inlet pipe; ionizing the gas by the discharge between the cathode electrode that is filamentous and heated by energization and the anode electrode provided around the cathode electrode; and emitting the ionized gas toward the surface of the substrate by accelerating the ionized gas inside the acceleration area.
